

SUPPLEMENTARY INFORMATION

Supplementary Methods:**Expression and purification of Sgs1 and sgs1-K706A**

We first modified the pFastBac-HTB vector (Invitrogen) by adding a FLAG tag after the pre-existing (His)₆ tag. A DNA fragment encoding full length Sgs1 protein with a carboxyl-terminal HA tag was fused in frame to the FLAG and His₆ tags in this vector, and a bacmid was generated in the *E. coli* strain DH10Bac (Invitrogen). The tagged Sgs1 protein was expressed in insect cells. All the purification steps were carried out at 0–4°C. The insect cell pellet (~15 g from 1 L culture) was resuspended in 100 ml of K buffer (20 mM KH₂PO₄, pH 7.4, 10% glycerol, 0.5 mM EDTA, 0.01% Igepal, 1 mM DTT) with a cocktail of protease inhibitors (aprotinin, chymostatin, leupeptin, and pepstatin A at 5 µg/ml each, and also 1 mM phenyl-methylsulfonyl fluoride) and 500 mM KCl. Cells were disrupted by sonication for 30 seconds, and the lysate was clarified by ultracentrifugation (100,000Xg for 45 min) and then incubated with 1 ml of anti-FLAG-M2 resin (Sigma) for 2 h. The matrix was washed once with 20 ml K buffer containing 500 mM KCl and 2 mM each of ATP and MgCl₂, then washed three additional times with 20 ml of K buffer containing 500 mM KCl. Sgs1 was eluted by incubating the matrix with 3 ml of K buffer containing 500 mM KCl and 200 µg/ml FLAG peptide (Sigma) for 1 h. The eluate was mixed gently with 300 µl of Nickel-NTA beads (Qiagen) for 1 h. The matrix was washed as before, except that 15 mM imidazole was included in the buffers. To elute Sgs1, 1 ml of K buffer containing 500 mM KCl and 200 mM imidazole was used. The eluate was filter-dialyzed into K buffer containing 500 mM KCl in an Ultracel-30K concentrator (Amicon) and further concentrated down to 200 µg/ml. The purified Sgs1 protein (25–40 µg) was divided into 2 µl aliquots, frozen in liquid nitrogen, and stored at -80°C. The integrity of the nearly homogeneous Sgs1 protein was confirmed by immunoblotting using anti-poly-His and anti-HA antibodies. Expression and purification of the sgs1-K706A mutant protein followed the same procedures, and with a similar yield.

Expression and purification of Dna2, dna2-D657A, dna2-K1080E and dna2-Δ405N

For yeast expression of Dna2, *pGAL10-DNA2* (gift from Judith Campbell), which expresses Dna2 with a FLAG and HA double tag at the N-terminus, a (His)₆ tag at the C-terminus, and under the control of the *GAL10* promoter, was introduced into the protease deficient yeast strain BJ5464 (*MATα ura3-52 trp1 leu2Δ1 his3Δ200 pep4::HIS3 prb1Δ1.6R can1 GAL*). An overnight culture was diluted 1:100 into 15 L of -Ura omission medium with 2% raffinose. Cells were cultured at 30°C until the OD₆₆₀ reached 0.8, at which time galactose was added to 2% to induce Dna2 expression. Cells were harvested by centrifugation after 12 h of further incubation, and the pellet (70 g) was stored frozen at -80°C. After grinding the yeast pellet with dry ice, 70 ml of ice-cold lysis buffer (40 mM KH₂PO₄, pH 7.4, 20% glycerol, 1 mM EDTA, 1% NP-40, 2 mM DTT, 600 mM KCl, and the cocktail of protease inhibitors as above) was added to the extract. All the subsequent steps were carried out at 0–4°C. The extract was clarified by ultracentrifugation (100,000Xg for 45 min) and then applied onto a 5 ml column of Nickel-NTA agarose (Qiagen). The column was washed with 20 ml K buffer with 500 mM KCl and 2 mM each of ATP and MgCl₂ and then with 60 ml of K buffer with 500 mM KCl and 10 mM imidazole. Dna2 was eluted with 20 ml of K buffer with 500 ml

KCl and 200 mM imidazole, and the eluate was mixed gently with 0.5 ml anti-FLAG-M2 resin for 2 h. After washing the matrix with 30 ml of K buffer with 500 mM KCl and 0.1% NP-40 and 10 ml of K buffer with 500 mM KCl, Dna2 was eluted with 1 ml of K buffer plus 500 mM KCl and 200 µg/ml FLAG peptide for 1 h. The eluate that contained purified Dna2 (200 µg/ml) was dialyzed against K buffer with 500 mM KCl overnight before being concentrated using an Ultracel-30K concentrator (Amicon) to 2 mg/ml and stored at -80°C in 2 µl aliquots. The overall yield of Dna2 was ~200 µg.

To generate dna2-K1080E and dna2-D657A, point mutations were introduced into *pGAL10-DNA2* vector by site-directed mutagenesis and both mutants were expressed and purified similarly as wild type. The yield of dna2-D657A was lower than wild type (~40 µg from 15 L culture), while that of dna2-K1080E was similar as that of the wild type protein.

For dna2-Δ405N expression, the *pGAL10-DNA2* vector was used as template and the appropriate primer sets were employed in a PCR reaction. The amplified *dna2-Δ405N* gene with an added C-terminal (His)₆ tag was introduced into the pESC-URA vector to put the mutant *dna2* gene under the control of the *GAL10* promoter with a FLAG tag present at the N-terminus. The FLAG-(His)₆-tagged dna2-Δ405N mutant protein was purified from the BJ5464 strain with the same procedure as wild-type Dna2, but with a higher yield of ~800 µg from a 15 L culture.

Expression and purification of Top3-Rmi1 and top3-Y356F-Rmi1 complexes

Plasmid pLK79, which harbors Top3 with C-terminal (His)₆ and V5 tags in pET11c, was a gift from Steven Brill. The *Rmi1* protein-coding sequence was PCR amplified from a Rmi1 expression construct, pNJ7138 (a kind gift from Dr. Steven Brill), and cloned into the NdeI and XhoI sites of pRSF-Duet1 to generate pRSF-Duet-Rmi1. For Top3-Rmi1 expression, pLK79 and pRSF-Duet-Rmi1 were co-transformed into the BL21 Rosetta strain. An overnight culture was diluted 1:50 into 10 L 2X LB medium (supplemented with 50 mg/L Ampicillin, 30 mg/L Kanamycin and 37 mg/L Chloramphenicol), and when the culture reached OD₆₀₀ = 0.8, 0.2 mM IPTG was added to induce Top3 and Rmi1 expression. Following a 12h incubation at 16°C, cells were harvested by centrifugation to yield a 50 g cell pellet. For Top3-Rmi1 purification, the cell pellet was resuspended in 200 ml of K buffer with 150 mM KCl and lysed by sonication (1 min, three times). The extract was clarified by ultracentrifugation (100,000Xg for 90 min) and loaded onto a 5 ml SP Sepharose column. After washing the column with 100 ml K buffer plus 150 mM KCl, it was developed with a 100 ml linear gradient from 150 to 575 mM KCl. The Top3-Rmi1 complex was eluted at ~350 mM KCl, and the peak fractions of which were pooled and incubated with 0.5 ml of Nickel-NTA beads for 2 h. The matrix was then washed once with 10 ml K buffer plus 250 mM KCl and 1 mM each of ATP and MgCl₂, three times with 10 ml K buffer plus 250 mM KCl and 15 mM imidazole, before elution with K buffer with 250 mM KCl and 200 mM imidazole. The eluate was dialyzed against K buffer with 250 mM KCl, concentrated to 1 mg/ml in an Ultracel-30K concentrator (Amicon), and stored at -80°C in 2 µl aliquots. The yield of highly purified Top3-Rmi1 complex ~200 µg.

The Y356F point mutation was introduced into pLK79 by site-directed mutagenesis, and top3-Y356F was co-expressed with Rmi1, as above. The top3-Y356F-Rmi1 complex was purified as above, with a similar yield.

Expression and purification of Sae2

E. coli Rosetta cells (Novagen) harboring *pMAL-SAE2* that expresses MBP-Sae2-(His)₆ were grown at 37°C to OD₆₀₀ of 0.6 and induced for protein expression with 0.5 mM IPTG for 14 h at 16°C. Cells (30 g from 20 L culture) were resuspended in 100 ml of cell breakage buffer (50 mM Tris HCl, pH 7.5, 10% sucrose, 10 mM EDTA, 1 M KCl, 0.01% Igepal, 1 mM 2-mercaptoethanol, and the cocktail of protease inhibitors as above) and lysed by sonication. The lysate was clarified by ultracentrifugation and incubated with 5 ml Ni-NTA (Qiagen) for 1.5 h at 4°C. Then, the beads were washed with 25 ml each of buffer K containing 1 M KCl and 300 mM KCl, respectively, and the bound proteins were eluted with buffer K containing 200 mM imidazole and 300 mM KCl. The eluate was diluted with 3 volumes of buffer K and applied onto a SP-Sepharose column (8 ml), which was developed with a 80 ml gradient of 10-500 mM KCl in buffer K. Peak fractions containing MBP-Sae2 were pooled and incubated with 2.5 ml amylose beads (NEB) for 1 hr. The beads were washed with 15 ml each of buffer K containing 300 mM KCl and 100 mM KCl, respectively, and the bound proteins were eluted with 10 mM maltose in buffer K containing 100 mM KCl. The eluate was further fractionated in a 1 ml Mono Q column, using a 30 ml gradient of 100-400 mM KCl in buffer K. The peak fractions were pooled, concentrated, and stored at -80°C.

Other proteins

Mre11-Rad50-Xrs2 and mre11-3-Rad50-Xrs2 complexes were purified from yeast cells as described before¹. Yeast RPA was purified as described². Human RPA was purified as described³. *E. coli* SSB was purchased from New England Biolabs.

DNA substrates for helicase and nuclease assays

φX174 replicative form DNA (~90% supercoiled) (New England Biolabs) was linearized by digestion with PstI. To construct the radiolabeled DNA substrates, various oligonucleotides were 5'-labeled with [γ-³²P] ATP using T4 polynucleotide kinase and then annealed to their partner oligonucleotides by heating an equimolar amount of the oligonucleotides at 95°C for 10 min in buffer (50 mM Tris-HCl, pH 7.5, 10 mM MgCl₂, 100 mM NaCl), followed by slow cooling to room temperature. Hybridized DNA substrates were separated from unannealed oligonucleotides in a 10% polyacrylamide gel in TAE buffer and recovered from the gel by electro-elution in a dialysis tubing into TBE buffer at 4°C. The helicase substrate for Sgs1 was obtained by hybridizing oligos H1 and H2 (Supplementary Table 1), and that for Dna2 was obtained by hybridizing oligos H3 and H4 (Supplementary Table 1). The H3/H4 and H3/H5 hybrids were used to test the effect of single-strand DNA binding proteins on Dna2's 5' and 3' endonuclease activities, respectively, in Figure 6 and Supplementary Figure 10. Oligos H2, H6 and H7 were annealed to generate the 5' FLAP structure to test Dna2's endonuclease activity in Supplementary Figure 2.

Sgs1 ATPase and helicase assay

ATPase assay for Sgs1 was carried out at 30°C with 5 nM Sgs1 in 20 µl buffer (40 mM Tris-HCl, pH 7.5, 2 mM MgCl₂, 1 mM DTT, 100 µg/ml BSA, 50 mM KCl, 0.2 mM ATP, 25 nCi [γ -³²P]-ATP). Analysis by thin layer chromatography was as described⁴. For the helicase assay, Sgs1 (5, 10 and 20 nM) was incubated at 37°C with the radiolabeled DNA substrate (5 nM) in 10 µl of buffer (30 mM Tris-HCl, pH 7.5, 1 mM DTT, 100 µg/ml BSA, 2 mM MgCl₂, 2 mM ATP, and the ATP regenerating system). The reaction was stopped by treatment with SDS (0.2% final) and proteinase K (0.5 mg/ml) for 3 min at 30°C. The reaction mixtures were resolved in a 10% polyacrylamide gel in TAE buffer at 4°C. Gels were dried onto DE3 paper and then analyzed in a Personal Molecular Imager FX PhosphorImager (Bio-Rad).

Dna2 ATPase assay

Dna2 or mutant (20 nM each) was incubated with ϕ X174 ssDNA (33 µM nucleotides) in 10 µl buffer (40 mM Tris-HCl, pH 7.5, 2 mM MgCl₂, 1 mM DTT, 100 µg/ml BSA, 50 mM KCl, 0.2 mM ATP, 25 nCi [γ -³²P]-ATP) at 30°C for 5 min. Analysis was as described for Sgs1.

Dna2 nuclease assay

Dna2 or mutant (1, 2, and 5 nM) was incubated at 37°C for 10 min with the radiolabeled 5' FLAP DNA substrate (5 nM) in 10 µl buffer (25 mM Tris-HCl, pH 7.5, 1 mM DTT, 100 µg/ml BSA, 2 mM MgCl₂). The reaction mixtures were processed and analyzed, as above.

Dna2 helicase assay

The dna2-D657A or dna2-K1080E protein, 10 or 20 nM, was incubated at 37°C for 10 min with the radiolabeled DNA substrate (5 nM) in 10 µl buffer (25 mM Tris-HCl, pH 7.5, 2 mM ATP, 1 mM DTT, 1 mM MgCl₂, 100 µg/ml BSA). The reaction mixtures were processed and analyzed, as above.

Resection of non-radiolabeled linear DNA substrate

The reaction was conducted with 25 ng of either ϕ X174 linear (1.4 nM ends) or supercoiled DNA with 50 nM MRX, 50 nM Sgs1, 100 nM Dna2, 50 nM top3-Y356F-Rmi1 and 500 nM RPA for 2 min at 30°C. Analysis was done by agarose gel electrophoresis as above and ethidium bromide staining of the DNA species.

Double Holliday Junction dissolution assay

The dHJ substrate⁵ (1 nM) was incubated with Sgs1 (5 nM) and either Top3/Rmi1 (5 nM) or top3-Y356F/Rmi1 (5 nM) in buffer (40 mM Tris-HCl, pH 7.5, 50 mM KCl, 2 mM MgCl₂, 2 mM ATP, 1 mM DTT, 100 µg/ml BSA) at 30°C for 10 min. Analysis was as described⁵.

Preparation of pre-resected DNA

For making pre-resected DNA, a 50 µl reaction containing 3 nM internally ³²P-labeled blunt-ended DNA and 10 nM purified yeast Exo1 protein (a 5' to 3' exonuclease) was

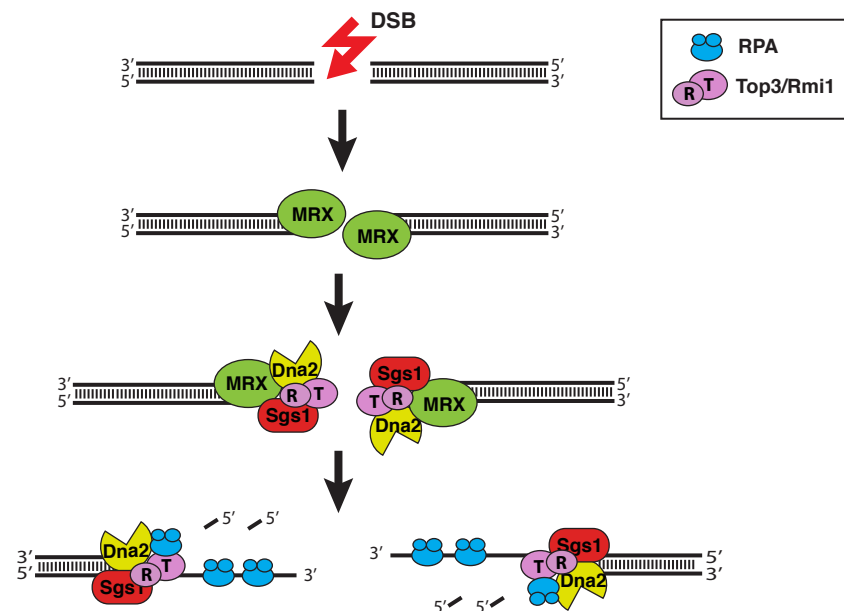
assembled in buffer (40 mM Tris-HCl, pH 7.5, 5 mM MgCl₂, 1 mM DTT, 100 ng/μl BSA) and incubated at 30°C for 10 min, followed by heating at 75°C for 5 min to inactivate Exo1. The Exo1 treatment resulted in ~30% release of the ³²P-label, as determined by phosphorimaging analysis of a 10% polyacrylamide gel in which a sample from the reaction mixture had been resolved. The pre-resected substrate was passed through a Biospin 6 (Bio-Rad) column to remove the bulk of the released ³²P-label. The control blunt ended substrate had been similarly incubated with heat-inactivated Exo1.

Mre11 nuclease assay

The DNA substrate (φX174 ssDNA or RFI DNA, 36 μM nucleotides) was incubated with Mre11 or mre11-3 (50 or 100 nM each) in 25 μl buffer (35 mM BisTris-HCl, pH 7.0, 1 mM DTT, 1 mM MnCl₂, 100 μg/ml BSA 50 mM KCl) at 37°C for 1 h. The reaction mixtures were deproteinized and then resolved in a 0.9% agarose gel in TAE buffer. The DNA species were stained with ethidium bromide and recorded in a gel documentation station. For testing the 3' exonuclease activity of Mre11 and mre11-3, 20 nM of either protein was used in a 25 μl reaction containing 3'-labeled linear dsDNA (3 kb) with 0.5 nM ends, 30 mM Bis-Tris HCl, pH 7.0, 1 mM DTT, 1 mM ATP, 100 ng/μl BSA, 50 mM KCl and the indicated amounts of Mg²⁺ and/or Mn²⁺. The reaction was incubated at 30°C for 15 min and stopped by 50 mM EDTA and 0.5 mg/ml of proteinase K. Analysis was by electrophoresis in a 10% TBE polyacrylamide gel and phosphorimaging of the dried gel.

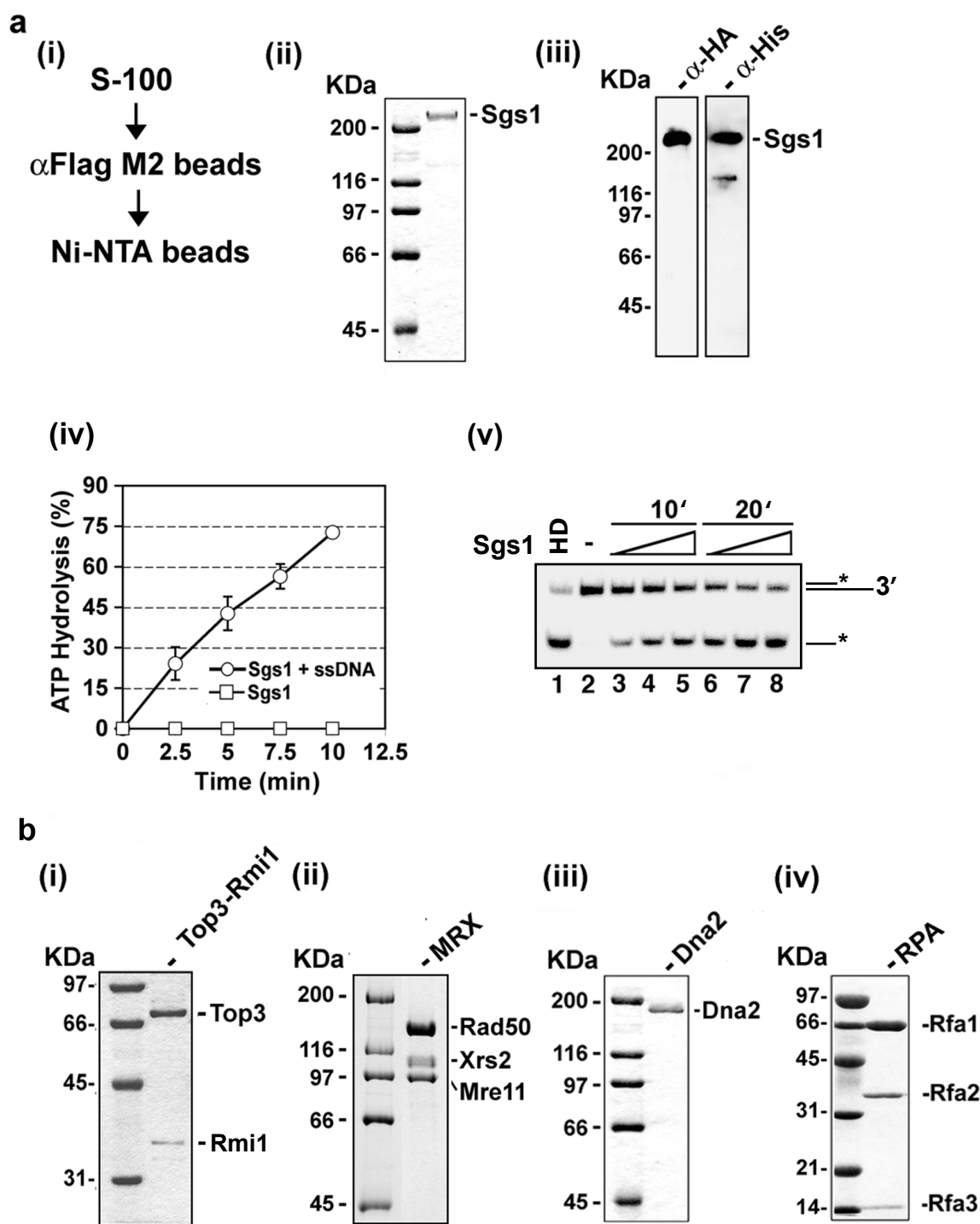
Affinity pulldown assay

To test for Dna2-RPA interaction, FLAG-(His)₆-tagged Dna2 or dna2-Δ405N (2 μg) and yeast RPA (2 μg), SSB (2 μg) or human RPA (2 μg) were incubated in 30 μl K buffer with 15 mM imidazole for 30 min at 4°C. After mixing with 10 μl Nickel-NTA resin (Qiagen) for 30 min at 4°C, the resin was washed three times with 200 μl of buffer, and proteins were eluted with 15 μl of 2% SDS. The supernatant (S), wash (W), and SDS eluate (E), 10 μl each, were analyzed by either 10 % SDS-PAGE for yeast RPA and human RPA or 4-20 % SDS-PAGE (Bio-Rad) for SSB.



Supplementary Figure 1. Model for the reconstituted DNA motor-driven end resection pathway.

MRX, by virtue of its high affinity for DNA ends, serves to recruit the resection machinery to the ends. The STR complex drives the separation of DNA strands, and Dna2, in conjunction with RPA, resects the 5' strand. RPA also sequesters the liberated 3' strand to prevent DNA re-annealing and protects the 3' strand from digestion.

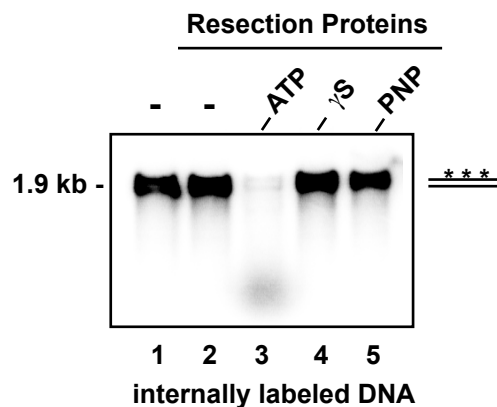


Supplementary Figure 2. Analyses of components of the DNA end resection machinery.

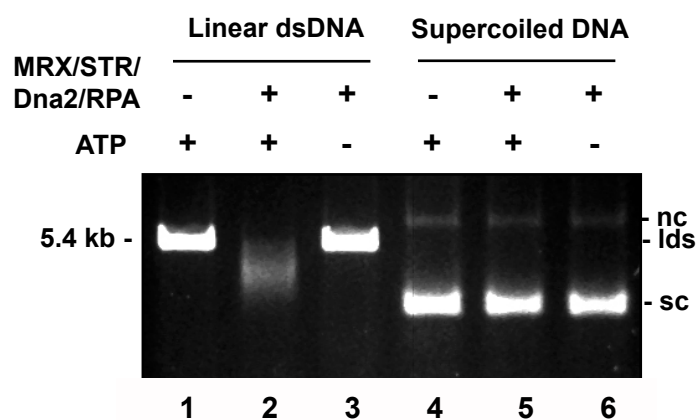
a, Schematic of Sgs1 purification (i). Sgs1, 500 ng, was analyzed by 7.5% SDS-PAGE (ii). Purified Sgs1, 100 ng, was immunoblotted with either anti-HA (left) or anti-polyhistidine (right) antibodies (iii). ATPase activity of Sgs1 was examined without DNA or with ssDNA (iv). DNA helicase activity of Sgs1 was assayed (v). Lane 1 contained heat-denatured substrate (HD) and lane 2 was the reaction blank.

b, Top3-Rmi1 complex, 1 µg, was analyzed by 10% SDS-PAGE (i). Note that the under-representation of Rmi1 relative to Top3 likely stemmed from a slight proteolysis of Rmi1. Mre11-Rad50-Xrs2 complex, 3 µg, was analyzed by 8% SDS-PAGE (ii). Dna2, 1 µg, was analyzed by 8% SDS-PAGE (iii). RPA, 1.5 µg, was analyzed by 12% SDS-PAGE (iv).

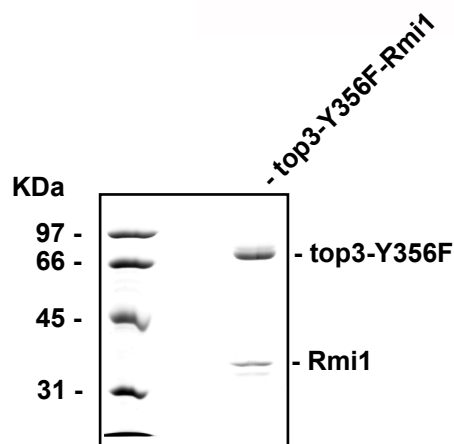
a



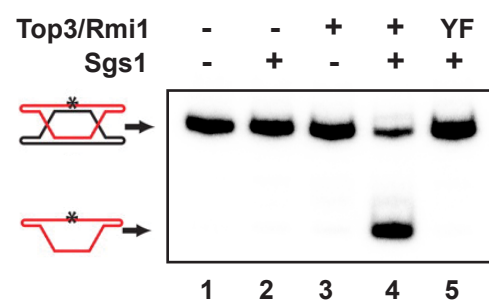
b



c



d



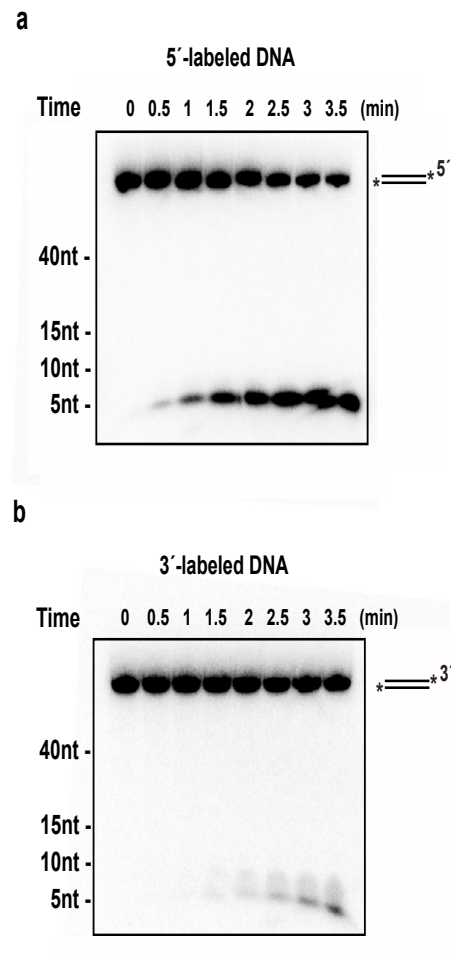
Supplementary Figure 3. DNA resection by MRX/Dna2/STR/RPA is ATP dependent and initiates from DNA ends

a, Internally labeled DNA was incubated with the reconstituted resection machinery in the absence or presence of ATP, ATPγS (gS) or AMP-PNP (PNP). The incubation time was 5 min. Lane 1 was the no protein control.

b, The use of linear dsDNA and supercoiled DNA revealed that resection initiates from DNA ends. Symbols: nc, nicked circular DNA; lds, linear duplex DNA; sc, supercoiled DNA.

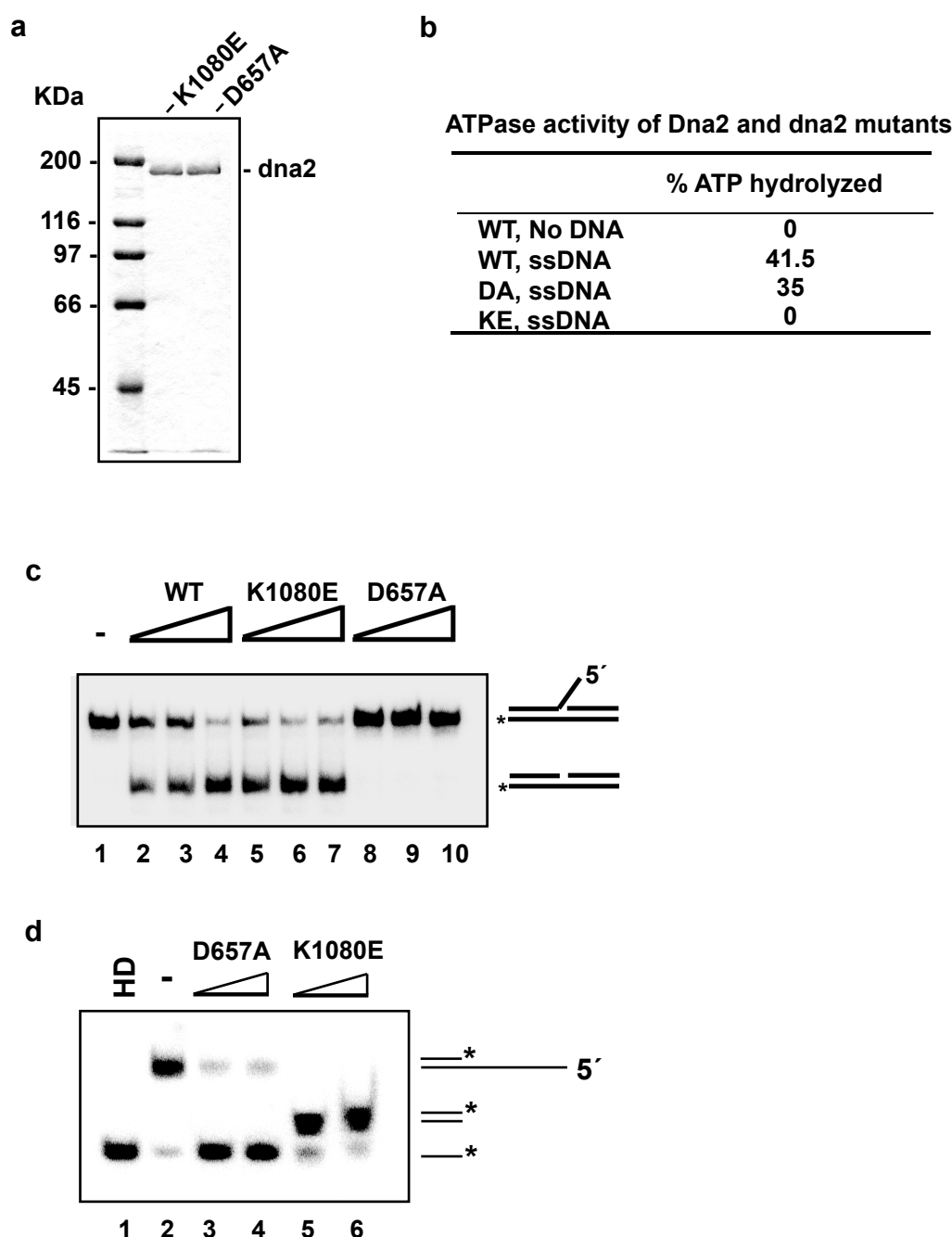
c, top3-Y356F-Rmi1 complex, 1 μg, was analyzed by 10% SDS-PAGE (i). Note that the under-representation of Rmi1 relative to top3-Y356F likely stemmed from a slight proteolysis of Rmi1.

d, Top3/Rmi1 and top3-Y356F/Rmi1 were examined together with Sgs1 for the ability to dissolve a synthetic double Holliday junction substrate⁵



Supplementary Figure 4. Time course of end resection with 5'- or 3'-labeled DNA.

a, b, Samples taken at different time points during the end resection reaction that used 5'- or 3'-labeled substrate were resolved in a denaturing polyacrylamide gel followed by phosphorimaging analysis.



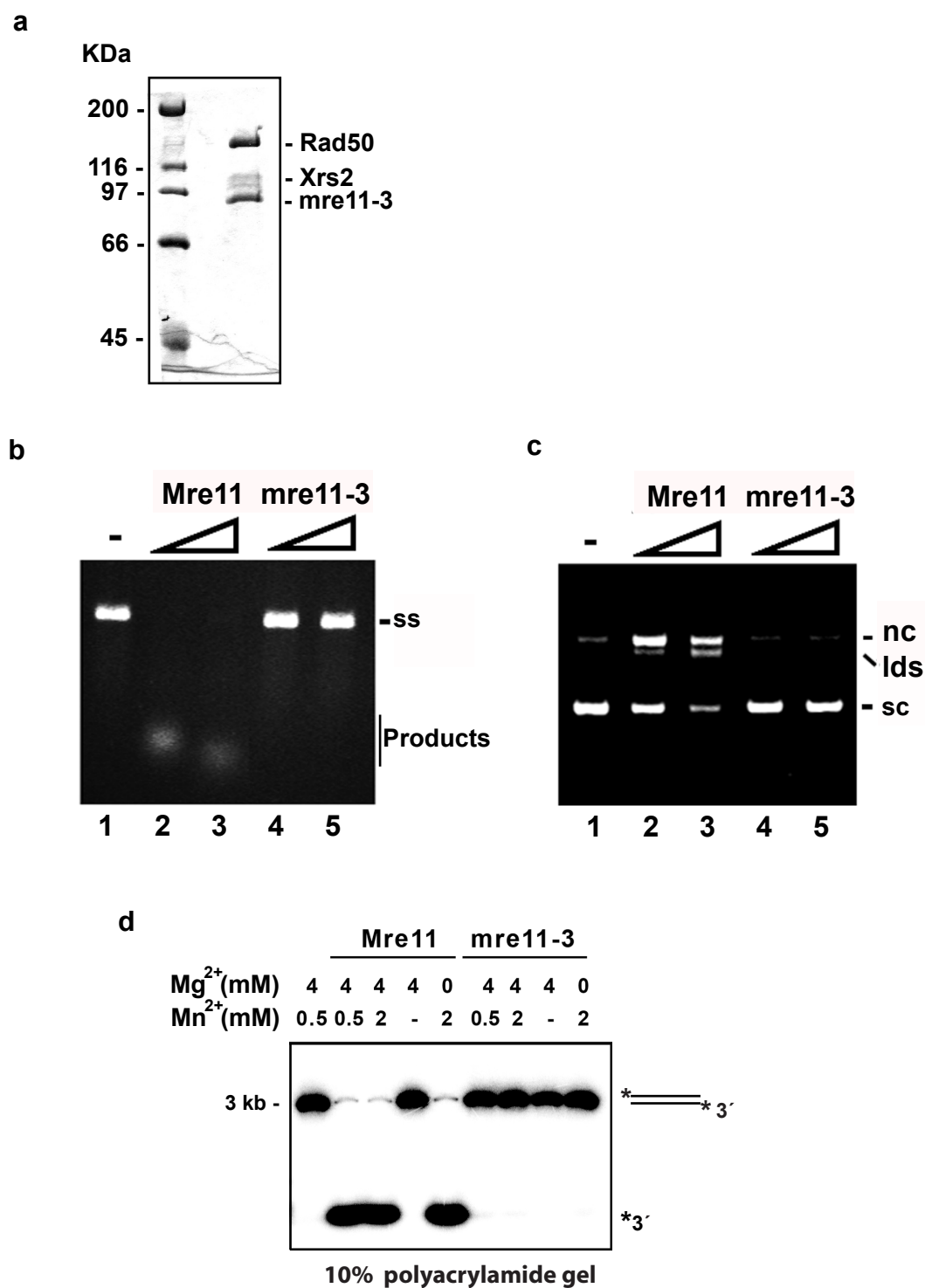
Supplementary Figure 5. Characterization of wild-type Dna2 and the dna2-K1080E or dna2-D657A mutant.

a, Purified dna2-K1080E and dna2-D657A proteins, 500 ng each, were analyzed by 7.5% SDS-PAGE.

b, ATPase activity of Dna2 was examined without DNA or with ssDNA, while the activity of dna2-K1080E or dna2-D657A was examined only in the presence of ssDNA.

c, The endonuclease activity of Dna2 (WT), dna2-K1080E (K1080E) or dna2-D657A (D657A) was examined. Lane 1 was the no protein control.

d, The dna2-D657A (D657A) and dna2-K1080E (K1080E) were examined for their ability to unwind or digest a DNA substrate with a 5' ssDNA overhang. Lane 1 contained heat-denatured substrate (HD) and lane 2 was the no protein control. As expected, dna2-D657A unwound the substrate but did not digest it, whereas dna2-K1080E produced a nucleolytic product only. With wild type Dna2, the nuclease activity predominated over the helicase activity under the conditions used (data not shown).

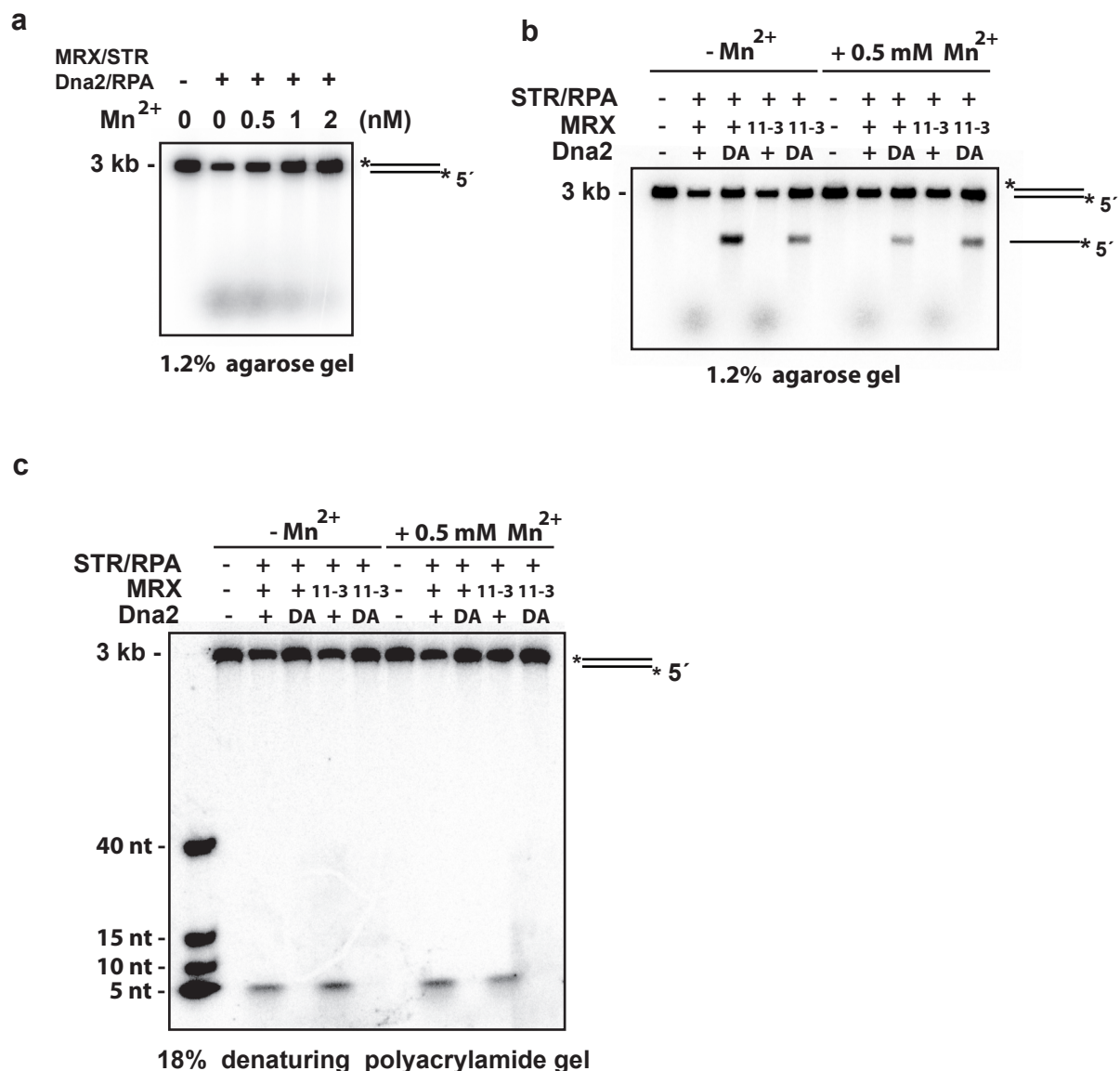


Supplementary Figure 6. Analyses of mutant MRX harboring mre11-3.

a, The purified MRX mutant complex that harbored mre11-3, 3 µg, was analyzed by 8% SDS-PAGE.

b, c, Mre11 and mre11-3 were examined for nuclease activity. Lane 1 was the no protein control. Symbols: ss, circular ssDNA; nc, nicked circular DNA; lds, linear duplex DNA; sc, supercoiled DNA.

d, Mre11 and mre11-3 were examined for 3' exonuclease activity with 3'-labeled linear duplex DNA in the presence of the indicated amounts of Mg²⁺ and/or Mn²⁺.

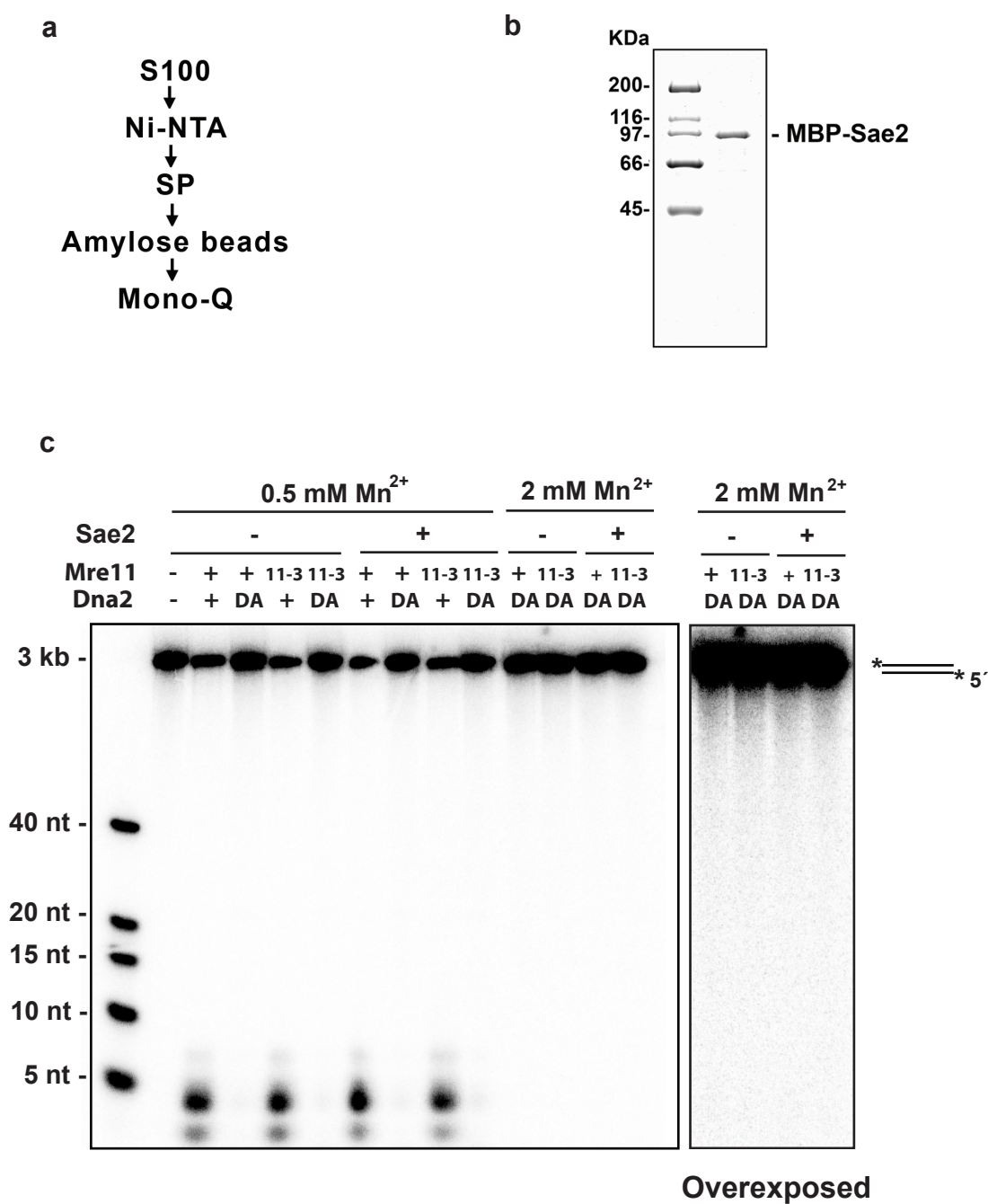


Supplementary Figure 7. Initial 5' digestion depends on Dna2 but not Mre11.

a, Resection reactions were conducted with 5'-labeled DNA and increasing concentrations of Mn²⁺. The incubation time was 2 min and the reactions were analyzed in a 1.2% agarose gel.

b, Combinations of Dna2 or dna2-D657A (DA) mutant and/or MRX or mutant MRX harboring mre11-3 (11-3) were tested in conjunction with STR and RPA with the 5'-labeled DNA substrate in the absence or presence of 0.5 mM Mn²⁺. The incubation time was 2 min and the reactions were resolved in a 1.2% agarose gel followed by phosphorimaging analysis.

c, The same reactions in **b** were resolved in a denaturing polyacrylamide gel followed by phosphorimaging analysis to examine the product size.

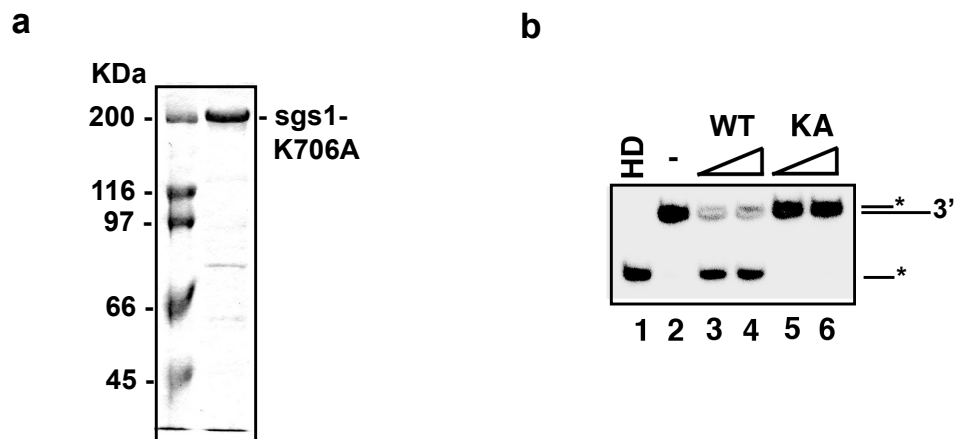


Supplementary Figure 8. Sae2 does not alter Dna2-dependent 5' cleavage.

a, Purification scheme for recombinant MBP-Sae2-(His)₆.

b, Purified Sae2, 1 µg, was analyzed by 10% SDS-PAGE.

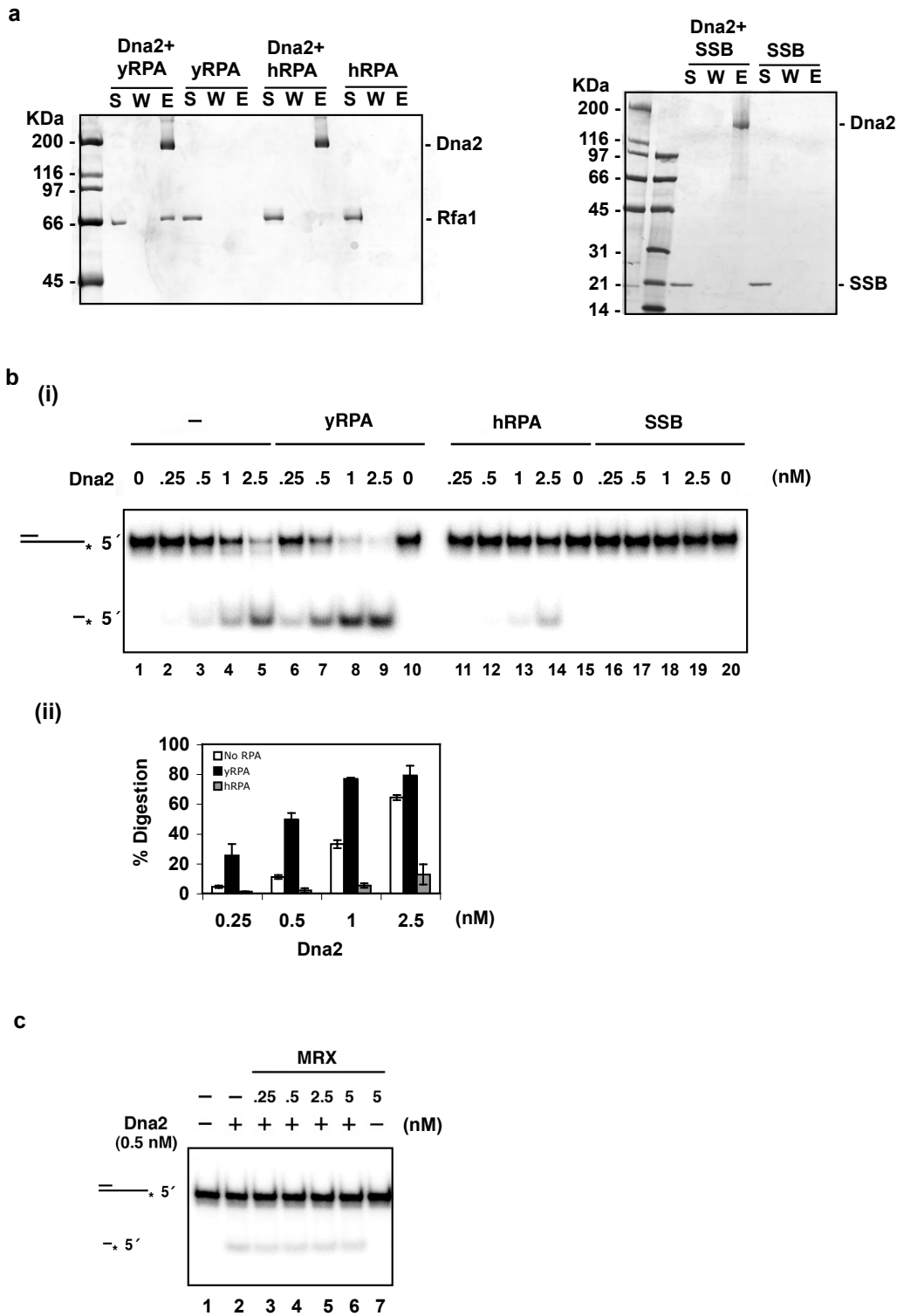
c, Resection reactions were assembled with either 0.5 mM or 2 mM Mn²⁺, incubated and analyzed as in Supplementary Figure 6. Sae2 (10 nM) was included in the reaction, where indicated. An over-exposure of the last four reactions that contained 2 mM Mn²⁺ is also shown.



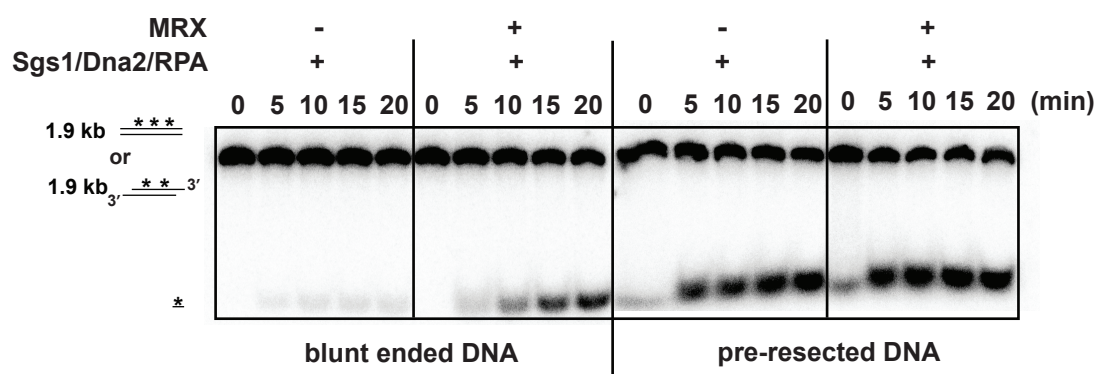
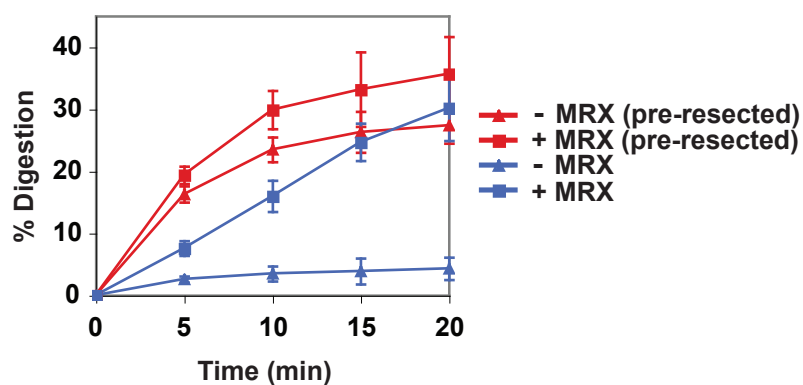
Supplementary Figure 9. Characterization of the sgs1-K706A mutant.

a, Purified sgs1-K706A, 1 μ g, was analyzed by 7.5% SDS-PAGE.

b, Sgs1 (WT; 10 and 20 nM) and sgs1-K706A (KA; 20 and 40 nM) were tested for helicase activity. Lane 1 contained heat-denatured (HD) substrate and lane 2 was the no protein control.



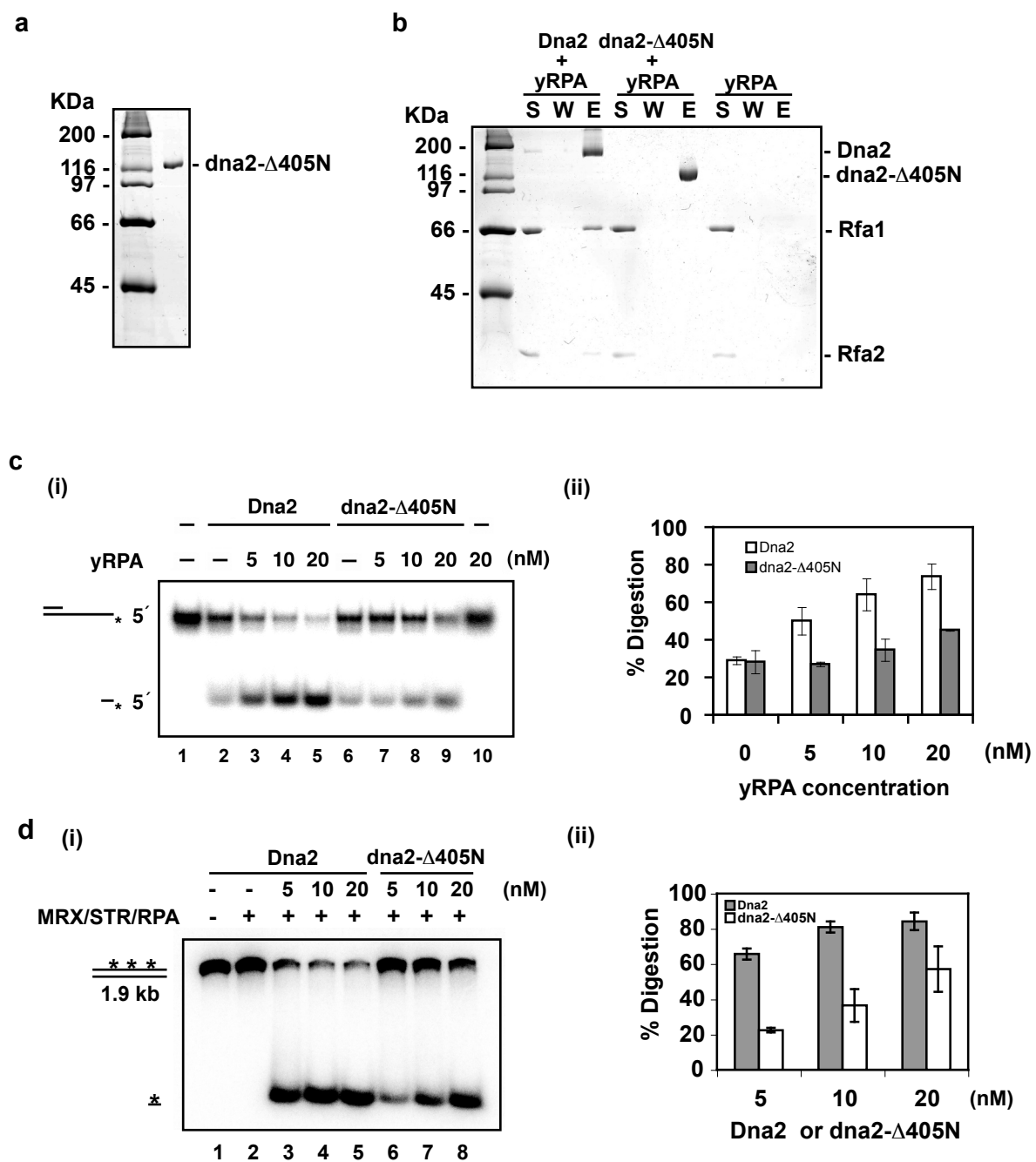
Supplementary Figure 10. Species-specific interaction and enhancement of Dna2 by RPA.
a, Affinity pulldown with Nickel-NTA resin to examine the interaction of (His)₆-tagged Dna2 with yRPA, hRPA, or *E. coli* SSB. The supernatant (S), wash (W) and SDS eluate (E) fractions were analyzed by SDS-PAGE.
b, The effect of yRPA, hRPA, and *E. coli* SSB (20 nM each) on the ability of Dna2 to digest a DNA substrate with a 5' ssDNA overhang was tested (i) and the results were quantified and plotted (ii).
c, The effect of MRX on the 5' DNA incision activity of Dna2 (0.5 nM) was tested.

a**b**

Supplementary Figure 11. Resection of DNA harboring 3' ssDNA tails by Sgs1-Dna2-RPA is independent on MRX.

a, The effect of MRX on end resection mediated by Sgs1-Dna2-RPA was analyzed with either blunt-ended or pre-resected uniformly labeled DNA (0.3 nM ends) as described in Figure 3a.

b, Mean values $\pm$ SD from three independent experiments were plotted.



Supplementary Figure 12. Characterization of dna2-Δ405N.

a, Purified dna2-Δ405N, 500 ng, was analyzed by 7.5% SDS-PAGE.

b, Affinity pulldown with Nickel-NTA resin to examine the interaction of (His)₆-tagged Dna2 and dna2-Δ405N with yRPA. The supernatant (S), wash (W) and SDS eluate (E) fractions were analyzed by SDS-PAGE.

c, The ability of Dna2 and dna2-Δ405N (1 nM each) to digest a 5' ssDNA overhang was examined in the presence of various amounts of RPA (i). The results were plotted (ii).

d, Dna2 and dna2-Δ405N were tested in the resection system with uniformly labeled DNA (i) and the results were plotted (ii).

Supplementary Table 1: Oligo nucleotides used for making DNA substrates

Oligo Name	Sequence
H1	ATTAAGCTCTAAGCCATGAATTCAAATGACCTCTTATCAA
H2	TTGATAAGAGGTCATTTGAATTCATGGCTTAGAGCTTAATTGCTGAATCTGGTGCTGG GATCCAACATGTTTTAAATATG
H3	ATGTCACATATTGAAGCGCTGATCACTGTCTCCATCGAACGTTGATAAGAGGTCATTTG AATTCATGGCTTAGAGCTTAAT
H4	ATTAAGCTCTAAGCCATGAA
H5	CAGCGCTTCAATAGTGACAT
H6	CATATTTAAACATGTTGGATCCCAGCACCAGATTCAGCATACGTTACCGATCGTACG TTCGATGCTGGCTACTGCTAGC
H7	GCTAGCAGTAGCCAGCATCGAACGTACGATCGGTAACGTA

- 1 Chen, L., Trujillo, K., Ramos, W., Sung, P. & Tomkinson, A. E. Promotion of Dnl4-catalyzed DNA end-joining by the Rad50/Mre11/Xrs2 and Hdf1/Hdf2 complexes. *Mol. Cell* **8**, 1105-1115 (2001).
- 2 Van Komen, S., Macris, M., Sehorn, M. G. & Sung, P. Purification and assays of *Saccharomyces cerevisiae* homologous recombination proteins. *Methods Enzymol.* **408**, 445-463 (2006).
- 3 Sigurdsson, S., Trujillo, K., Song, B., Stratton, S. & Sung, P. Basis for avid homologous DNA strand exchange by human Rad51 and RPA. *J. Biol. Chem.* **276**, 8798-8806 (2001).
- 4 Petukhova, G., Stratton, S. & Sung, P. Catalysis of homologous DNA pairing by yeast Rad51 and Rad54 proteins. *Nature* **393**, 91-94 (1998).
- 5 Wu, L. & Hickson, I. D. The Bloom's syndrome helicase suppresses crossing over during homologous recombination. *Nature* **426**, 870-874 (2003).